Smoke and Strife: Fire Exposure and Social Conflict in Indonesia

Hariaksha Gunda

The University of Alabama

October 10, 2026

Abstract

Seasonal land and forest fires blanket much of Indonesia in severe air pollution ("haze") each year. I ask whether short-run exposure to this fire-driven pollution affects the incidence of social conflict. Because local fire activity may itself respond to local economic and political conditions, I instrument a district's smoke exposure with fire activity upwind of it, isolating variation driven by wind-borne smoke transport rather than by local conditions. I find no effect of haze on conflict on average, but a negative effect in districts where conflict is already common. This effect is concentrated in riots and violence against civilians rather than in peaceful protests or organized armed conflict, and only the effect on riots and violence against civilians survives the full set of robustness checks; the broader effects on all conflict events and on political violence lose significance at the 5% level once spatial correlation in the errors is accounted for. I find no effect on conflict-related fatalities. The direction of the effect runs counter to the climate–conflict and pollution–crime literatures, which typically find that adverse environmental shocks raise conflict. The affected events are committed both by crowds (riots) and by armed actors, most often state security forces (violence against civilians). This points to an acute, short-run channel, such as reduced outdoor activity or reduced reporting during severe haze, rather than to the intensified grievances through which environmental shocks are usually thought to affect conflict.

1 Introduction

A large literature links climate variation to conflict, typically through channels such as agricultural income shocks, resource scarcity, or temperature-driven aggression (Hsiang et al.,

2013). Indonesia offers a distinctive setting: recurring, large-scale peat and forest fires, driven by land-clearing practices and intensified during El Niño-induced dry seasons, generate transboundary haze events that degrade air quality across much of the western archipelago. Unlike slow-moving climatic shifts, fire-driven haze is a salient, high-frequency, and geographically variable shock that likely affects daily life: outdoor mobility, school and market operations, public health, and, the focus of this paper, the propensity for public gatherings that can escalate into social and political conflict.

This paper studies whether exposure to fire-driven air pollution causally affects the incidence of social conflict at the district-month level. Answering this question is complicated by the fact that local fire activity may be endogenous to local conflict: land disputes, weak governance, and economic distress that drive conflict may also drive land-clearing fires, and unobserved local shocks may affect both. I address this with an instrumental-variables design that exploits the directional, physical nature of smoke transport. For each district-month, I construct an instrument equal to the total fire radiative power (FRP) of fire detections that lie *upwind* of the district, that is, in locations whose smoke plumes the prevailing wind would carry toward the district, within a fixed radius and angular cone defined by the wind direction. Upwind fire activity affects a district's local pollution exposure through atmospheric transport but is plausibly uncorrelated with the district's own local economic and political shocks, conditional on district and time fixed effects.

I construct this instrument at a finer spatial resolution than prior applications of this design to Indonesia, using GADM Level-2 administrative boundaries (kabupaten/kota, $n = 447$ in the conflict-matched sample) rather than provinces, and link fire detections (NASA VIIRS), wind fields (ERA5 reanalysis), and conflict events (ACLED) via point-in-polygon spatial joins. This yields a district-month panel of 55,875 observations spanning February 2015 to June 2025.

The paper's central finding is a contrast between the full-sample and a conflict-active subsample. In the full panel, instrumented fire intensity (log FRP) has essentially zero effect on conflict counts, a null that survives a Poisson control-function specification designed for severely zero-inflated count data (following Lu and Yamazaki, 2023). However, this null masks substantial heterogeneity: in districts where conflict is not a rare event (defined as districts with conflict activity in at least 30% of sample months), a one-log-point increase in local fire intensity reduces conflict events by 0.11 events per district-month ($p = 0.005$) and political-violence events by 0.046 ($p = 0.006$). Splitting this restricted-sample effect by event type shows that it operates entirely through riots and violence against civilians (-0.046, $p = 0.002$), with no detectable effect on battles or remote violence ($p = 0.964$). A finer decomposition into individual ACLED event types (Section 5.5) shows that peaceful

protests, by far the largest category and the one most tied to planned public assembly, do not respond, while riots and *violence against civilians* respond about equally, so the mechanism appears narrower, and less well understood, than a simple disruption of planned public assembly. I develop this qualification in Sections 5.5 and 6.

The remainder of the paper proceeds as follows. Section 2 relates this paper to three literatures: climate and conflict, wind- and pollution-instrumented designs linking air quality to behavior, and Indonesia-specific haze research. Section 3 describes the data and panel construction. Section 4 describes the instrument and empirical strategy. Section 5 presents the paper's central results: the full-sample null, the conflict-active heterogeneity analysis, and a two-category and a finer four-category event-type decomposition, along with brief summaries of several additional robustness and specification checks reported in full in the Appendix. Section 6 discusses interpretation and limitations, Section 7 outlines future research directions, Section 8 concludes, and the Appendix reports additional data description, preliminary correlational evidence motivating the instrumented design, and the complete robustness checks summarized in Section 5.

2 Related Literature

This paper draws on three distinct literatures: economic studies of climate and conflict, a more recent literature using wind- and pollution-based instruments to link air quality to violence and crime, and research specifically documenting the severity of Indonesian fire-driven haze.

2.1 Climate and conflict

Economists have documented a robust link between climate variability and conflict. Hsiang et al. (2013) synthesize 60 quantitative studies spanning interpersonal violence to civil war and institutional collapse, and find that each one-standard-deviation shift toward warmer temperatures or more extreme rainfall increases the frequency of interpersonal violence by 4% and intergroup conflict by 14% (median estimates, $p < 0.001$). Burke et al. (2015), surveying a related but distinct set of 55 studies using a hierarchical meta-analysis, report a similar pattern: a one-standard-deviation increase in contemporaneous temperature raises interpersonal conflict by 2.4% and intergroup conflict by 11.3%, with an additional two-period cumulative effect of rainfall on intergroup conflict of 3.5% per standard deviation.

Two studies in this literature are especially relevant to the Indonesian setting studied here. Hsiang et al. (2011) show that the El Niño/Southern Oscillation (ENSO) is associ-

ated with global civil-conflict risk: in countries whose climate is strongly coupled to ENSO ("teleconnected" countries), annual conflict-onset risk rises from 3% in La Niña years to 6% in El Niño years, implicating ENSO in roughly 21% of civil conflicts worldwide between 1950 and 2004. Because Indonesian fire activity is itself strongly modulated by ENSO, as discussed further below, this result is a useful benchmark for the El Niño-phase heterogeneity check proposed in Section 7. Miguel et al. (2004) provide the canonical template for using a plausibly exogenous weather variable to instrument for an economic shock and then testing its effect on conflict: using rainfall variation as an instrument for income growth in 41 African countries (1981–99), they find that a five-percentage-point negative growth shock increases the likelihood of civil conflict by one-half the following year, an effect that is not significantly different in richer, more democratic, or more ethnically diverse countries. The wind-instrumented design developed in Section 4 follows the same logic of using an exogenous environmental variable to isolate variation in an otherwise endogenous treatment, applied here to a different environmental shock (wind-transported smoke) and setting.

2.2 Pollution and behavior: wind-instrumented designs

A more recent line of work shows that air pollution itself, independent of any economic-shock channel, causally affects aggressive and criminal behavior, using designs methodologically close to the one developed in this paper. Herrnstadt et al. (2021) compare violent crime on the upwind and downwind sides of major Chicago interstates on days when wind blows orthogonally to the road, finding that violent crime rises by 1.9% on the downwind side, with no effect on property crime; the effect is largest at moderate temperatures, when people are more likely to be outdoors, and is contemporaneous rather than lagged. Bondy et al. (2020) instrument daily air-pollution exposure in London with both atmospheric inversions and wind direction, finding that a 10-point increase in the Air Quality Index raises the crime rate by 2.6% in their preferred wind-instrumented specification, an effect they attribute to a rational-choice mechanism in which pollution increases the discounting of future punishment. Burkhardt et al. (2019) use county-by-day variation in PM2.5 and ozone across the United States and find that a 10% increase in PM2.5 raises violent crime by 0.14% and a 10% increase in ozone raises it by 0.3%, with effects concentrated in assaults and no detectable effect on property crime; they interpret pollution as a contemporaneous irritant that spurs impulsive aggression.

These three studies all find that short-run pollution exposure *increases* violence, through either an aggression/irritant channel or a rational-choice discounting channel. This paper's finding, that haze *decreases* riots and violence against civilians, is therefore not a simple

extension of this literature; if anything, it cuts against the modal finding that short-run pollution exposure raises aggressive behavior, a contrast discussed further in Section 6.

2.3 Indonesian haze as a pollution shock

A third body of work documents the severity of Indonesian fire-driven haze as a pollution shock in its own right. Jayachandran (2009) studies the 1997 Indonesian fires, mostly intentionally set by logging and palm-oil companies to clear land and which burned out of control because of unusually dry, windy, El Niño-driven conditions, and finds that the resulting smoke caused 15,600 "missing" fetal, infant, and child deaths, a 1.2% reduction in the affected birth cohorts, with effects concentrated in third-trimester in utero exposure and twice as large in below-median-consumption districts. Marlier et al. (2013) link Indonesian and southeast Asian fire activity directly to ENSO, showing that regional fire carbon emissions were roughly 50 times higher during the 1997 El Niño than the 2000 La Niña (1,069 versus 21 Tg C), and estimate that fire emissions during strong El Niño years raise regional adult cardiovascular mortality by approximately 2%, or roughly 10,800 deaths annually.

Sheldon and Sankaran (2017) instrument Singaporean air-pollution levels with Indonesian fire radiative power (FRP), the same measure used as the treatment variable in this paper, and find that a one-standard-deviation increase in fire activity raises Singaporean polyclinic attendance for acute respiratory infections and conjunctivitis by 0.7 standard deviations. They report being unable to construct a wind-direction instrument because of a lack of consistent Singaporean wind data, and instead aggregate FRP across all Indonesian coordinates. The upwind instrument developed in Section 4 directly extends this approach, adding the directional refinement that Sheldon and Sankaran (2017) were unable to implement.

2.4 The closest precedent

Most closely related to the present paper is Lu and Yamazaki (2023), which studies a different Indonesian natural-resource channel, marine fish catches, and its relationship to conflict, using a search-by-radius spatial design across 757 ocean-and-land cells. They find that both industrial and non-industrial fish catches are positively associated with conflict events, that illegal, unreported, and unregulated catches are more likely than legal catches to cause violent conflict, and that protests and riots are the event types most sensitive to the resource shock, while fighting and terrorism are the least sensitive. The event-type results in this paper differ in instructive ways. Here the effect is negative rather than positive, it is concentrated in riots and violence against civilians, and protests do not respond at all (Section 5.5). Lu and Yamazaki (2023) pool protests, riots, and strategic developments into a single category,

so their results cannot say whether protests and riots respond differently. The two papers agree on one point: battles and remote violence respond least, and here not at all. Beyond this substantive comparison, Lu and Yamazaki (2023) is also the methodological source for the Poisson IV control-function approach summarized in Section 5 and reported in full in Appendix Section C.1, used to address the extreme zero-inflation of the conflict outcomes studied here.

3 Data

All data-construction and analysis code for this project, including the panel-construction and spatial-matching steps described below, is available at this link.[1]

3.1 Fire activity

Fire detections come from the NASA VIIRS active fire product (375m resolution, J1/Suomi-NPP, Nov. 2000–Feb. 2026 archive, restricted to Indonesia with a 0km buffer). I retain only detections classified as vegetation fires (`type == 0`) with high or nominal confidence (`confidence` $\in \{h, n\}$), dropping volcanic and industrial heat sources and low-confidence detections. Each retained detection carries a precise latitude, longitude, date, and fire radiative power (FRP, in megawatts), which serves as the measure of fire intensity. Detections are assigned to districts via a point-in-polygon spatial join against GADM v4.1 Level-2 boundaries, and aggregated to total monthly FRP per district, FRP_{dt}. The treatment variable is $\log(1 + \text{FRP}_{dt})$, denoted logFRP_{dt}.

3.2 Wind

Wind fields come from ERA5 reanalysis (Hersbach et al., 2023), providing monthly mean 10-meter eastward (u_{10}) and northward (v_{10}) wind components on a $0.25^\circ \times 0.25^\circ$ grid. I compute wind speed and the meteorological wind direction (*the direction the wind blows from*, measured clockwise from north) at each grid cell. Each grid cell is assigned to the district whose polygon contains it via a spatial join; for districts smaller than the grid resolution (no grid cell falls inside the polygon), the nearest grid cell to the district centroid is used instead. District-month wind direction and speed are computed as the average of u_{10} and v_{10} over all grid cells assigned to that district.

[1] `https://github.com/Hariaksha/climate-conflict`

3.3 Conflict

Conflict event data come from the Armed Conflict Location & Event Data Project (ACLED) for Indonesia. ACLED's `admin2` field reports district names using English translations (e.g., "East Jakarta"), while GADM uses Indonesian district names (e.g., "Jakarta Timur"). I construct a crosswalk between the two naming systems via (i) direct string matching, (ii) a manual override dictionary for known renames and translation idiosyncrasies (e.g., "Yogyakarta" → "Kota Yogyakarta"), and (iii) directional-word translation rules (e.g., "East X" → "X Timur", "X Islands" → "Kepulauan X"). This crosswalk matches 449 of the 469 district names in ACLED. The ACLED extract runs from 1 January 2015 to 10 June 2025; the analysis window begins in February 2015 because the first month is lost to the one-month lag. Of the 21,653 events ACLED records in this window, 21,347 (98.6%) are matched to a district. The 306 dropped events comprise 126 in 14 regencies created by administrative splits (*pemekaran*) that are absent from the GADM 4.1 boundary set, 120 with no district recorded in ACLED (113 of them in Jakarta), and 60 in six districts that exist in GADM under a different spelling but are missed by the crosswalk (e.g., Binjai, Tanjung Balai, Banjar). These 21,347 *matched* events (as opposed to *all* recorded events, mapped in Appendix Figure A1) are the basis for the district-month conflict counts used in the regressions and for all event-type shares reported below.

I construct two outcome variables at the district-month level:

- **events**: count of all ACLED events of any type.
- **pv_events**: count of "political violence" events, defined as ACLED event types *Battles*, *Violence against civilians*, *Riots*, and *Explosions/Remote violence* (excluding peaceful protests and strategic developments).

For the event-type decomposition (Section 5.4), I further split political-violence events into two mutually exclusive categories: **Riots/VAC** (ACLED types *Riots* and *Violence against civilians*, VAC) and **Battles/Explosions** (ACLED types *Battles* and *Explosions/Remote violence*). The two categories sum to the political-violence count. Neither includes ACLED's *Protests* category, which Section 5.5 examines separately.

3.4 Panel construction

The analysis panel is the cross-join of all 447 conflict-matched districts and all 125 months from February 2015 to June 2025, yielding 55,875 district-month observations. Appendix Table A1 reports summary statistics. The conflict outcomes are heavily zero-inflated: 82.5%

of district-months record zero events of any type, and 95.8% record zero political-violence events.

3.5 Geographic distribution

Appendix Figures A1 and A2 map the raw conflict and fire data underlying the panel; the conflict map shows *all* recorded events in the sample window, of which the 98.6% matched to a district enter the analysis. Conflict events are concentrated in Java, Sumatra, and Papua, while fire activity is concentrated in the peatland and forest regions of Sumatra, Kalimantan, and Papua – areas that are spatially distinct from, but geographically adjacent to, the most conflict-active districts, consistent with the smoke-transport mechanism the upwind instrument is designed to exploit.

4 Empirical Strategy

Before developing the instrumented design, I verified that a purely correlational, non-instrumented approach does not reveal a stable relationship between local fire intensity and conflict at several spatial and temporal resolutions; full results are reported in Appendix Section B and motivate the identification strategy developed in this section.

4.1 The upwind instrument

The key empirical challenge is that local fire intensity, logFRP_{dt}, may be correlated with unobserved local shocks that also drive conflict (e.g., land disputes, weak local enforcement, economic distress). I address this with an instrument based on the physical transport of smoke: for each district d and month t, I define

$$Z_{dt} = \sum_{f \in \mathcal{U}_{dt}} \text{FRP}_f,$$

where $\mathcal{U}_{dt}$ is the set of individual VIIRS fire detections f in month t that (i) fall within a fixed radius of district d's centroid, and (ii) lie within a fixed angular cone of the direction from which district d's wind is blowing in month t. Intuitively, Z_{dt}, the *upwind instrument*, measures fire activity in locations whose smoke plumes the wind would carry toward district d, but which are themselves located elsewhere and thus not subject to district d's local shocks. I compute Z_{dt} efficiently using a `scipy.spatial.cKDTree` spatial index over fire detection coordinates, querying for detections within the radius of each district centroid and filtering by the angular criterion. The instrument enters the regressions as $\text{logZ}_{dt} = \log(1 + Z_{dt})$.

4.2 First stage

The first-stage regression is

$$\text{logFRP}_{dt} = \alpha_d + \gamma_t + \delta_0 \,\text{logZ}_{dt} + \delta_1 \,\text{logZ}_{d,t-1} + \eta_{dt}, \tag{1}$$

where α_d and γ_t are district and year-month fixed effects, and standard errors are clustered at the province level (38 provinces) to account for spatial correlation in wind patterns within province. Table 1 reports the results. Both the contemporaneous and one-month-lagged upwind instrument are strong, positive predictors of local fire intensity. Because the errors are clustered, I assess instrument strength with the Montiel Olea and Pflueger (2013) effective F-statistic, which is robust to heteroscedasticity, autocorrelation, and clustering; the conventional homoscedastic first-stage F greatly overstates instrument strength in this setting. The effective F is 85.6 for the first stage in Table 1 and 82.6 for the first stage of the distributed-lag IV specifications below, which additionally conditions on $\text{logFRP}_{d,t-1}$. Both comfortably exceed the Staiger and Stock (1997) rule of thumb of 10 and the conservative Montiel Olea and Pflueger (2013) 5% critical value for rejecting the hypothesis that the 2SLS bias exceeds 10% of the OLS benchmark (23.1).

Table 1: First stage: instrument relevance

	Dependent variable: logFRP_{dt}	
	Coefficient	SE
logZ_{dt} (upwind FRP, contemporaneous)	0.304^{***}	(0.032)
$\text{logZ}_{d,t-1}$ (upwind FRP, lag 1)	0.037^{***}	(0.011)
District FE	Yes	
Year-month FE	Yes	
R^2 (within)	0.137	
Effective F (Montiel Olea and Pflueger, 2013)	85.6	
Observations	55,875	

Notes: Standard errors clustered by province (38 clusters). The effective F is computed from the province-clustered variance of the first-stage coefficients. The first stage of the distributed-lag IV specifications (IV-1, IV-2), which also includes $\text{logFRP}_{d,t-1}$, has effective $F = 82.6$; that of the contemporaneous-only specifications (IV-3, IV-4) has effective $F = 90.4$. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

4.3 Second stage

The structural equation of interest is

$$y_{dt} = \alpha_d + \gamma_t + \beta_0 \,\text{logFRP}_{dt} + \beta_1 \,\text{logFRP}_{d,t-1} + \varepsilon_{dt}, \tag{2}$$

where $y_{dt} \in \{\text{events}_{dt}, \text{pv_events}_{dt}\}$, and logFRP_{dt} is instrumented by $(\text{logZ}_{dt}, \text{logZ}_{d,t-1})$ while $\text{logFRP}_{d,t-1}$ enters as an exogenous control.[2] I estimate four specifications: (IV-1, IV-2) the distributed-lag model in Eq. (2) for `events` and `pv_events`, and (IV-3, IV-4) a contemporaneous-only version (no lag terms) for the same two outcomes. All specifications include district and year-month fixed effects, with standard errors clustered by province.

5 Results

5.1 Full-sample IV estimates

Table 2 reports the full-sample IV estimates. Across all four specifications, the coefficient on instrumented logFRP_{dt} is small in magnitude and statistically indistinguishable from zero (all $p > 0.2$). This is despite the very strong first stage reported above, so the null is not attributable to weak-instrument bias.

[2]The estimator (`pyfixest`'s `feols` with an IV specification) supports a single endogenous regressor; I therefore instrument the contemporaneous treatment while including its lag as a control, following standard practice in distributed-lag IV designs. Appendix Section C.4 relaxes this by instrumenting both the contemporaneous treatment and its lag simultaneously; the main conflict-active and event-type results are robust to this change, though it reveals a significant effect on `pv_events` in the full panel that the partially-instrumented specification does not detect.

Table 2: Second-stage IV estimates, full panel (447 districts, 55,875 obs.)

	IV-1 events (dist. lag)	IV-2 pv_events (dist. lag)	IV-3 events (contemp.)	IV-4 pv_events (contemp.)
logFRP_{dt}	0.0008	−0.0021	0.0074	0.0011
	(0.0066)	(0.0021)	(0.0060)	(0.0016)
	$[p = 0.906]$	$[p = 0.314]$	$[p = 0.226]$	$[p = 0.504]$
$\text{logFRP}_{d,t-1}$	0.0003	0.0017	–	–
	(0.0046)	(0.0011)		
	$[p = 0.947]$	$[p = 0.138]$		
District FE	Yes	Yes	Yes	Yes
Year-month FE	Yes	Yes	Yes	Yes
Observations	55,875	55,875	55,875	55,875

Notes: logFRP_{dt} instrumented by upwind instrument (contemporaneous and one-month lag in IV-1/IV-2; contemporaneous only in IV-3/IV-4). Standard errors clustered by province in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

5.2 Zero-inflation and the full-sample null

A natural concern is that the full-sample null reflects the extreme zero-inflation of the conflict outcomes (82.5% zeros for `events`, 95.8% for `pv_events`; Appendix Table A1). In the majority of district-months, conflict is structurally absent regardless of local fire intensity (e.g., in remote or sparsely populated districts with no history of organized political activity), so a linear model estimated on the pooled sample may be dominated by observations in which the outcome simply cannot respond to the treatment. I pursue two complementary strategies to address this: (i) restricting estimation to "conflict-active" districts (Section 5.3), and (ii) a Poisson control-function model designed for count data (Appendix Section C.1).

5.3 Heterogeneity: conflict-active districts

I re-estimate the IV-1/IV-2 specifications on subsamples of districts with at least one conflict event in at least $\tau \in \{5\%, 10\%, 20\%, 30\%\}$ of the 125 sample months. This restriction preserves the district-month panel structure and lag design but concentrates estimation on districts where conflict is not a near-deterministic zero. Table 3 reports the results.

Table 3: IV estimates by conflict-activity threshold

Threshold τ	Districts	Obs.	Zero share	β(events)	β(pv_events)
$\geq 5\%$	323	40,375	76.8%	-0.0090 (0.0106)	-0.0040 (0.0026)
				$[p = 0.402]$	$[p = 0.140]$
$\geq 10\%$	242	30,250	71.5%	-0.0119 (0.0139)	-0.0063 (0.0039)
				$[p = 0.399]$	$[p = 0.116]$
$\geq 20\%$	132	16,500	59.3%	-0.0476^{*} (0.0271)	-0.0218^{**} (0.0100)
				$[p = 0.089]$	$[p = 0.036]$
$\geq 30\%$	86	10,750	50.3%	-0.1117^{***} (0.0369)	-0.0458^{***} (0.0156)
				$[p = 0.005]$	$[p = 0.006]$

Notes: Each row restricts the sample to districts with at least one conflict event in $\geq \tau$ of the 125 sample months, and re-estimates the IV-1/IV-2 distributed-lag specifications (coefficient on contemporaneous logFRP_{dt} reported; lag coefficients omitted). Effective first-stage F-statistics (Montiel Olea and Pflueger, 2013) range from 46.8 (at $\tau = 20\%$) to 68.8 (at $\tau = 5\%$), and equal 48.3 at $\tau = 30\%$, all above the conservative 5% critical value of 23.1. Standard errors clustered by province in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

The pattern is unambiguous: as the sample is restricted to progressively more conflict-active districts, the point estimate becomes more negative and more precisely estimated, becoming statistically significant at conventional levels for $\tau \geq 20\%$ and highly significant at $\tau \geq 30\%$. At the strictest threshold (86 of the 447 districts, with the remaining 361 districts dropped, have conflict activity in at least 30% of months[3]), a one-log-point increase in local fire intensity is associated with a reduction of 0.112 conflict events and 0.046 political-violence events per district-month, economically large effects relative to the conditional mean of conflict in these more active districts. The instrument remains strong throughout (effective $F \geq 46.8$), so this pattern is not an artifact of weakening instrument power in smaller subsamples.[4]

Because the 86-district subsample is selected on the conflict outcome itself, the resulting estimate should be interpreted as a *local* effect specific to districts where conflict is a recurring phenomenon, not as the population-average effect of fire exposure on conflict in Indonesia

[3]The 86 retained districts are concentrated in a handful of provinces: Jawa Timur (14 districts), Papua (12), Jawa Barat (11), Jakarta Raya (5), and Jawa Tengah (4) account for 46 of the 86; the remaining 40 districts are spread across 28 other provinces, with 1–3 districts each.

[4]This is the primary specification reported throughout the paper. Appendix Section C.3 shows that the all-events estimate at this threshold loses statistical significance under an additional province × calendar-month fixed effect controlling for shared regional seasonal fire cycles, and that under Conley spatial-correlation standard errors the all-events estimate is significant only at the 10% level ($p \approx 0.08$) and the political-violence estimate is not significant ($p \approx 0.13$), though the point estimates are essentially unchanged under both checks. The Riots/VAC result in Table 4 is robust to both checks. See Appendix Section C.3 for details.

(which Table 2 shows to be null). The monotonic strengthening of the effect across all four thresholds, rather than significance appearing only at one arbitrarily chosen cutoff, supports interpreting this as a true gradient rather than an artifact of a single specification search.

5.4 Event-type decomposition

To distinguish between possible mechanisms underlying the negative effect in conflict-active districts, I decompose the $\tau \geq 30\%$ sample's conflict outcome by ACLED event type into **Riots/VAC** (13.1% of all matched events) and **Battles/Explosions** (3.2% of all matched events), and re-estimate the IV specification for each category. Table 4 reports the results.

Table 4: Event-type decomposition, $\tau \geq 30\%$ conflict-active sample

	Riots/VAC	Battles/Explosions
logFRP_{dt}	-0.0462^{***}	0.0003
	(0.0134)	(0.0068)
	$[p = 0.002]$	$[p = 0.964]$
District FE	Yes	Yes
Year-month FE	Yes	Yes
Observations	10,750	10,750

Notes: *Riots/VAC* comprises ACLED event types *Riots* and *Violence against civilians*; *Battles/Explosions* comprises *Battles* and *Explosions/Remote violence*. Neither includes ACLED's *Protests* category (Table 5). Standard errors clustered by province in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

The negative effect on political violence identified in Table 3 is concentrated entirely in riots and violence against civilians: a one-log-point increase in fire intensity reduces Riots/VAC events by 0.046 ($p = 0.002$), essentially the entire political-violence effect (-0.046), while the effect on battles and remote violence is a precisely-estimated zero (0.0003, $p = 0.964$). Battles and remote violence, by contrast, appear insensitive to ambient air quality. Which specific event types drive the Riots/VAC effect, and whether peaceful protests, which neither category includes, respond as well, is addressed by the finer decomposition in the next subsection.

5.5 A finer decomposition: literal ACLED categories

The *Riots/VAC* category in Table 4 is a composite of two ACLED event types, *Riots* and *Violence against civilians*; it does not include ACLED's *Protests* category, the single largest

event type in the matched data (80.8% of all matched events) and the one most directly tied to planned, public assembly. If haze suppresses conflict by disrupting public assembly, this category should show the clearest negative effect of all. This section re-estimates the IV specification separately for four literal, non-overlapping ACLED categories: *Protests*, *Riots*, *Violence against civilians*, and *Strategic developments* (the last included as an additional placebo-style category, since strategic developments, e.g. arrests, recruitment, and looting, do not require public assembly and would not be expected to respond if haze worked by disrupting public assembly). Table 5 reports both the partially-instrumented specification (matching Table 4) and the fully-instrumented specification of Appendix Section C.4.

Table 5: Four-way event-type decomposition, literal ACLED categories, $\tau \geq 30\%$ sample

	Protests	Riots	Violence against civilians	Strategic developments
Partially instrumented				
logFRP_{dt}	-0.0544 $[p = 0.266]$	-0.0231^{*} $[p = 0.067]$	-0.0231^{**} $[p = 0.021]$	-0.0114^{*} $[p = 0.075]$
Fully instrumented				
logFRP_{dt}	-0.0463 $[p = 0.302]$	-0.0318^{***} $[p = 0.007]$	-0.0336^{***} $[p = 0.003]$	-0.0071^{*} $[p = 0.085]$
$\text{logFRP}_{d,t-1}$	-0.0172 $[p = 0.708]$	0.0256^{**} $[p = 0.031]$	0.0310^{***} $[p = 0.001]$	-0.0051 $[p = 0.416]$

Notes: Each column is a literal, non-overlapping ACLED event type (verified directly against the raw event-type field), not a composite category. "Partially instrumented" matches the specification in Table 4; "fully instrumented" matches Appendix Section C.4. Standard errors clustered by province. $N = 10{,}750$ for all columns. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

This decomposition complicates, rather than confirms, the disruption-of-assembly interpretation. *Protests*, the category most directly tied to planned public assembly and the largest by far, shows no statistically significant effect in either specification ($p = 0.27$–0.30). Within the Riots/VAC composite, the effect is split evenly between *Riots* and, more surprisingly, *Violence against civilians* (-0.023 each),[5] a category that does not obviously require public assembly in the way a riot or protest does; both are significant at the 1% level under full instrumentation and both display the same positive, significant one-month-lagged rebound documented in Appendix Section C.4. *Strategic developments*, included as a placebo-style category that should show no effect under either the original or the revised mechanism, instead shows a marginally significant negative effect ($p = 0.075$–0.085), weaker than Riots or Violence against civilians but not the precise null that *Battles/Explosions* shows in Table 4.

[5]The two categories involve very different actors. In the $\tau \geq 30\%$ sample, every *Riots* event is perpetrated by rioters (56.4% violent demonstrations, 43.6% mob violence), whereas *Violence against civilians* is perpetrated by state forces (60.0%), rebel groups (23.1%, chiefly the West Papua National Liberation Army), and political militias (16.0%), according to ACLED's `inter1` and `sub_event_type` fields.

On balance, these results do not support a mechanism specifically about disruption of *planned* public assembly, since Protests, the clearest example of such an event, is unaffected. They point instead to violent events specifically, whether committed by crowds (Riots) or by armed actors, most often state security forces, against civilians (Violence against civilians), though the weak effect on Strategic developments is not fully explained by this narrower story either, and could partly reflect reduced observability or reporting of lower-salience events during severe haze episodes rather than a purely behavioral channel. I do not have direct evidence to distinguish between these explanations with the data at hand, and I revise the interpretation in Section 6 accordingly: the mechanism underlying the conflict-active heterogeneity result is real, but it is not well described as "disruption of public assembly."

5.6 Additional robustness and specification checks

I subject the results above to four additional checks. Each is summarized briefly here and reported in full, with complete tables and discussion, in Appendix Section C.

Poisson IV control function. Following Lu and Yamazaki (2023), I re-estimate the full-sample relationship using a Poisson IV control-function approach (Terza et al., 2008) designed for severely zero-inflated count data. The control-function residual is statistically insignificant for both `events` ($p = 0.242$) and `pv_events` ($p = 0.374$), indicating minimal first-stage endogeneity bias and confirming that the full-sample null in Table 2 is not an artifact of the linear specification (Appendix Table A3).

Conflict-related fatalities. I do not find robust evidence that fire exposure affects conflict-related fatalities. Linear IV estimates are consistently negative but imprecise ($p \in [0.156, 0.189]$), and the one specification that yields a precisely-estimated effect (a Poisson control function) shows clear signs of residual endogeneity and relies on a severely restricted, non-representative sample (Appendix Table A4).

Exclusion restriction and inference. I implement a placebo test that replaces each district's upwind instrument with a randomly-assigned, implausibly distant (≥ 500 km) district's instrument, and find it is small, wrong-signed, and statistically insignificant, supporting the exclusion restriction. I also add a province $\times$ calendar-month fixed effect to absorb shared regional seasonal fire cycles, and compute Conley spatial-correlation standard errors that allow residual correlation across province boundaries. Under these stricter checks, the Riots/VAC result (Table 4) remains statistically significant throughout, while the broader all-events and political-violence results in Table 3 are more sensitive: the all-events result

loses significance under the added seasonal fixed effect, and neither all-events nor political violence is significant at the 5% level under Conley standard errors (all-events remains significant at the 10% level) (Appendix Tables A5–A7).

Fully-instrumented distributed-lag IV. The main specifications instrument only the contemporaneous treatment because `pyfixest` supports a single endogenous regressor, leaving the one-month lag as an uninstrumented control. Instrumenting both simultaneously leaves the conflict-active and event-type results unchanged in sign and, if anything, strengthens them, but reveals that the full-panel null is not uniform: political-violence events show a significant negative effect once the lag is properly instrumented (-0.0064, $p = 0.017$), alongside a new, economically large, positive lagged effect suggestive of a short-run disruption followed by a partial rebound (Appendix Table A8).

6 Discussion

The upwind-fire instrument is strong throughout (effective $F \approx 83$ in the full panel), so the design has ample power to detect downstream effects of fire-driven exposure if they exist, and the null found across the full population of 447 districts is a genuine average null rather than a symptom of weak identification: it holds under the baseline linear specification and survives the Poisson IV control-function check as well. That average null, however, conceals real heterogeneity. Among the subset of districts where conflict is not a rare event, fire intensity has a sizeable, negative effect on conflict events and on political violence specifically, and this heterogeneous effect does not hold up equally well across outcomes once spatial correlation in the errors is taken seriously: the Riots/VAC result withstands both Conley standard errors and an added province $\times$ calendar-month fixed effect (Appendix Section C.3), whereas the broader all-events and political-violence-event results in Table 3 do not, losing significance at the 5% level under Conley standard errors and, in the case of all-events, losing significance once seasonal fixed effects are added. Fully instrumenting the distributed-lag specification (Appendix Section C.4) reinforces the conflict-active result, ruling out the single-endogenous-regressor simplification used elsewhere as its source, but it also complicates the full-panel story: properly instrumenting the lag uncovers a significant negative contemporaneous effect on political-violence events even in the full panel, together with a new, economically large, positive lagged effect consistent with a short-run disruption followed by a partial rebound. Most importantly for interpretation, the finer event-type decomposition in Section 5.5 reshapes the mechanism itself: the effect is not concentrated in Protests, the ACLED category most directly tied to planned public assembly, but in-

stead in Riots and, less predictably, Violence against civilians, alongside a weaker effect on Strategic developments that a simple disruption-of-assembly account does not predict. Read together, these results support the Riots/VAC finding as real, but with a narrower and less well-understood mechanism than a simple disruption of public assembly, treat the broader conflict-active heterogeneity in Table 3 as informative but less conclusive evidence of the same pattern, and treat the full-panel null itself as only a partial null that does not extend cleanly to political-violence events once the lag structure is fully instrumented.

Interpretation. The pattern documented here is more nuanced than a straightforward disruption-of-collective-action story. The two-category decomposition (Table 4) confines the effect to riots and violence against civilians, with no effect on battles or remote violence. However, the finer decomposition in Section 5.5 shows that *Protests*, the event type most directly defined by planned public assembly, does not respond, while *Violence against civilians*, which does not require assembly in the same sense, responds as strongly as *Riots*. Because most violence against civilians in these districts is committed by state security forces and armed groups rather than by crowds, the common thread is not crowd behavior but violent political events, whether carried out by rioters or by armed actors. One possibility is that severe haze reduces the outdoor activity of both groups; another, which I cannot fully rule out, is that part of the pattern, particularly the weaker effect on *Strategic developments*, reflects lighter media or ACLED coverage of minor events when haze is severe rather than a purely behavioral response. Whatever its precise channel, the direction of the effect is the opposite sign from much of the climate-conflict literature, which typically finds that adverse environmental shocks *increase* conflict through resource scarcity or income-shock channels (Hsiang et al., 2013; Burke et al., 2015; Miguel et al., 2004). The contrast may reflect the very different time horizon at play: fire-driven haze is an acute, short-duration shock operating through immediate behavioral and logistical channels (crowds, security forces, and armed groups are less able or willing to act outdoors, or are less visible to reporters), rather than a persistent shock to income or resource availability that might generate grievance-based conflict over a longer horizon.

This result is also notable in light of the wind-instrumented pollution-and-behavior literature reviewed in Section 2. Herrnstadt et al. (2021), Bondy et al. (2020), and Burkhardt et al. (2019) all find that short-run air pollution *increases* violent crime, through either an aggression/irritant channel or a rational-choice discounting-of-punishment channel. The negative effect on riots and violence against civilians documented here runs in the opposite direction. One reconciliation is that these are different outcomes operating through different channels: the pollution-crime literature studies individual-level aggressive or criminal acts

that do not require coordination with others, whereas the ACLED events studied here are group actions, carried out by crowds in the case of riots and, by ACLED's definition, by organized armed groups, most often state security forces in this sample, in the case of violence against civilians. A mechanism that suppresses the willingness or ability to go outside could plausibly increase opportunistic, low-coordination offenses (consistent with the pollution-crime literature) while simultaneously decreasing group-based political violence (consistent with the result documented here) – these need not be in tension, even though the outcomes move in opposite directions.

Caveats and next steps. Several issues qualify the results above and merit continued attention. The conflict-active sample restriction raises an obvious external-validity concern: the documented effect applies to the subpopulation of districts where conflict is already a recurring phenomenon, so Section 5.3 is best read as a heterogeneity or local-effect analysis rather than a population-average one, with the full-sample null in Table 2 as the headline result – though Appendix Section C.4 shows even that null is not fully robust for `pv_events`. Inference is also uneven across outcomes: the placebo test in Appendix Section C.3 supports the exclusion restriction, but the shared-seasonality and Conley standard-error checks in the same section show that the all-events and political-violence results in Table 3 are less robust than the baseline specification implies, with `events` losing significance under the added province × calendar-month fixed effect and neither `events` nor `pv_events` significant at the 5% level under Conley standard errors. The Riots/VAC result in Table 4, which carries the paper's central mechanism claim, survives both checks and the fully-instrumented distributed lag (Appendix Section C.4), but Section 5.5 shows that even this result resists a simple disruption-of-planned-assembly explanation: Protests, the category most directly tied to that mechanism, is unaffected, and the effect instead runs through Riots and Violence against civilians. Given this, the four-category decomposition is the most informative and most transparent specific finding in the paper, and the broader event-count and two-category results are better treated as suggestive of a real but still poorly-understood mechanism than as conclusive evidence of disruption-of-assembly specifically. Two further loose ends deserve dedicated follow-up: the positive and often significant lagged coefficient that emerges once the lag is properly instrumented (Appendix Section C.4) was not anticipated by the original design and is only lightly treated here, and the marginally significant effect on Strategic developments (Section 5.5), a category not predicted to respond under any version of the mechanism considered, leaves open the possibility that part of the pattern reflects lighter media or ACLED coverage of minor events when haze is severe, rather than a purely behavioral response; separating these explanations would require external data on media

coverage or ACLED source density across high- and low-haze periods, which this analysis does not have.

7 Future Research Directions

This paper opens several avenues for future work, beyond the immediate robustness checks noted in Section 6.

The lagged rebound pattern. Appendix Section C.4 finds that once the one-month lag is properly instrumented rather than entering as an uninstrumented control, its coefficient becomes large, positive, and often statistically significant, alongside a negative contemporaneous effect. This pattern, a short-run disruption followed by a partial rebound the following month, was not anticipated by the original design and is reported here as a single robustness check rather than independently validated. A dedicated investigation, for example examining longer lag structures or testing whether displaced conflict activity specifically reappears in the same districts the following month, would clarify whether this is a real substitution dynamic or an artifact of this particular specification.

Distinguishing a behavioral channel from a reporting channel. Section 5.5 finds that the negative effect is concentrated in Riots and Violence against civilians, not in the literal Protests category, and that Strategic developments shows a weaker but still marginally significant effect. This pattern is consistent with at least two distinct explanations that the current data cannot separate: (i) haze reduces violent events, by rioters and armed actors alike, while leaving peaceful protest unaffected, or (ii) haze lowers the observability or media/ACLED reporting of minor events generally, mechanically depressing recorded counts across several event types regardless of any underlying behavioral channel. Distinguishing these would require external data on local media coverage, ACLED source density, or an independent measure of ground-truth event occurrence (e.g. satellite-detected nighttime gathering activity) during high- versus low-haze periods, none of which are used here.

Mechanism validation with finer-grained data. The disruption interpretation in Section 5.4 and Section 5.5 is inferred from the contrast across event-type categories, but is not directly tested. Daily (rather than monthly) conflict and pollution data, for example ground-level PM2.5 or satellite aerosol optical depth merged with daily ACLED event timing, would allow a sharper test of whether riot activity drops specifically on high-haze days,

and whether any displaced activity reappears once air quality improves (a substitution rather than elimination effect).

Heterogeneity by El Niño phase. Indonesian fire activity is strongly modulated by the El Niño–Southern Oscillation, with major haze episodes (e.g., 2015, 2019) occurring during El Niño years. Interacting the instrumented fire-intensity effect with ENSO phase would clarify whether the conflict-active-district results documented here are driven primarily by these extreme episodes or hold during more typical fire seasons as well.

Alternative and complementary outcomes. The analysis here focuses on conflict event counts. Incorporating economic outcomes (e.g., nighttime lights, mobility indices from cell-phone or satellite data, market activity) alongside conflict would help establish whether the disruption channel identified here extends to economic activity more broadly, and whether reduced economic activity itself partially explains the drop in riots and violence against civilians.

Spatial spillovers and displacement. The current design treats each district independently. If haze suppresses gatherings in one district, contention may be displaced to a neighboring, less-affected district rather than eliminated. A spatial model that allows for cross-district spillovers in the instrumented treatment, for example a spatial-lag or SAR-2SLS specification as in Lu and Yamazaki (2023), would test for such displacement and provide a more accurate estimate of the aggregate (rather than purely local) effect.

Extending the sample period and out-of-sample validation. The current panel runs through June 2025. As more recent VIIRS, ERA5, and ACLED data become available, extending the panel would increase power for the conflict-active subsample analysis and allow an out-of-sample test of whether the $\tau \geq 30\%$ threshold result documented in Table 3 holds in years not used to select that threshold.

Cross-country comparison. Indonesia is not the only country with recurring large-scale land-clearing fires (e.g., Brazil, parts of Sub-Saharan Africa). Replicating the upwind-instrument design in other settings would help establish whether the disruption-of-assembly mechanism identified here is a general feature of fire-driven air pollution shocks or specific to Indonesia's political and climatic context.

8 Conclusion

Using a novel district-month panel for Indonesia and an instrument based on the physical transport of wildfire smoke, I find no average effect of fire-driven pollution exposure on overall conflict under the baseline specification, but a negative effect among conflict-active districts. Only the effect on riots and violence against civilians survives the full set of robustness checks (Conley spatial standard errors and province × calendar-month fixed effects); the broader effects on all events and on political violence do not. Because the effect is concentrated in riots and in violence against civilians rather than in peaceful protests, it is better described as a reduction in violent political events, committed both by crowds (riots) and by armed actors, mostly state security forces (violence against civilians), than as a disruption of planned public assembly; the precise behavioral channel, and the possibility that it partly reflects reduced reporting of lower-salience events during severe haze, remains an open question. These results nonetheless suggest that short-run environmental shocks can affect conflict not only by altering incentives for violence, but by disrupting the day-to-day conditions under which violent political events occur or are observed.

This paper sits at the intersection of two literatures that, to date, have largely run in parallel: the economics literature on climate and conflict, which typically documents that adverse environmental shocks increase conflict through resource-scarcity or income channels (Hsiang et al., 2013; Burke et al., 2015), and a newer literature using wind- and pollution-based instruments to show that short-run air pollution increases individual aggressive and criminal behavior (Herrnstadt et al., 2021; Bondy et al., 2020; Burkhardt et al., 2019). By documenting a negative effect on group-based political violence, by crowds and armed actors alike, this paper suggests that the disruption channel through which environmental shocks affect conflict can run in the opposite direction from the individual-behavior findings in that second literature, a distinction that existing work has not, to my knowledge, explicitly tested.

Data Availability Statement

All analysis code, including the panel-construction, instrument-construction, and estimation notebooks described throughout this paper, is available at github.com/Hariaksha/climate-conflict. The conflict data (ACLED) and the processed wind data (ERA5) used in this paper are also hosted in that repository. The two largest raw inputs, fire detections and administrative boundaries, exceed standard GitHub file-size limits and are instead available directly from their original public sources:

- **Fire detections:** NASA VIIRS active fire products (Suomi-NPP and NOAA-20/J1,

375 m resolution), available via NASA FIRMS at `https://firms.modaps.eosdis.nasa.gov/download/`.

- **Administrative boundaries:** GADM v4.1 Indonesia boundaries (Level 2, kabupaten/kota), available at `https://gadm.org/download_country.html`.
- **Conflict events:** ACLED Indonesia data, available with registration at `https://acleddata.com`.
- **Wind fields:** ERA5 monthly averaged reanalysis on single levels (Hersbach et al., 2023), available via the Copernicus Climate Data Store at `https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels-monthly-means` (DOI: 10.24381/cds.f17050d7).

Replication of the full pipeline, from raw downloads through the figures and tables in this paper, requires retrieving the fire and boundary data from the sources above and placing them in the directory structure expected by the notebooks in the repository (documented in the repository's README). This paper contains modified Copernicus Climate Change Service information (2023); neither the European Commission nor ECMWF is responsible for any use that may be made of the Copernicus information or data it contains.

Appendix

This appendix reports additional data description (Appendix A), preliminary non-instrumented correlational evidence motivating the identification strategy in Section 4 (Appendix B), and the complete robustness and specification checks summarized in Section 5.6 (Appendix C).

A Additional Data Description

Table A1: Summary statistics, district-month panel (Feb. 2015–Jun. 2025)

Variable	Mean	SD	Zero share	N
$\log(1 + \text{FRP})$	2.149	2.260	–	55,875
events (all)	0.382	1.355	82.5%	55,875
pv_events (political violence)	0.062	0.406	95.8%	55,875

Notes: Panel covers 447 districts × 125 months. "Zero share" is the fraction of district-month observations with a value of exactly zero.

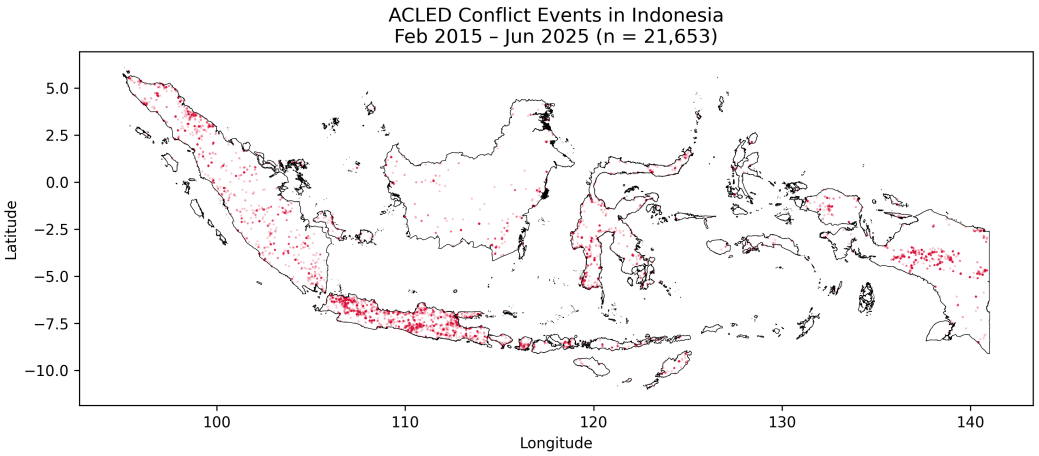


Figure A1: *All* ACLED conflict events recorded in Indonesia, February 2015–June 2025 ($n = 21{,}653$). The map shows every recorded event; the district-month panel uses the 98.6% (21,347) *matched* to a GADM Level-2 district (Section 3).

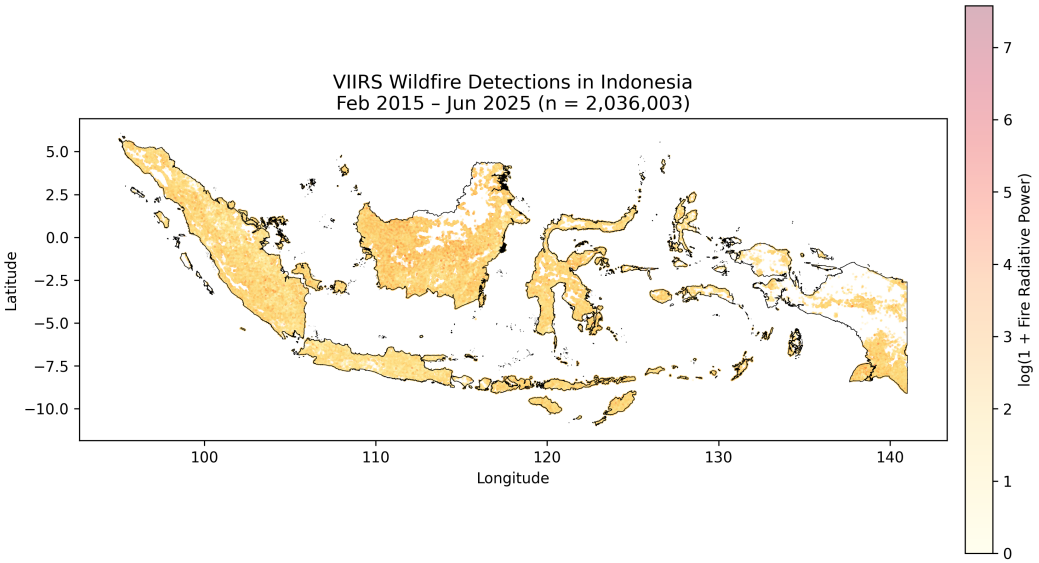


Figure A2: VIIRS wildfire detections in Indonesia, February 2015–June 2025 (n = 2,036,003; vegetation fires only, type=0, confidence h/n). Color indicates $\log(1 + \text{FRP})$.

B Preliminary Correlational Evidence

This appendix documents what a naive, non-instrumented approach finds, both to motivate the IV strategy used in the main text (Section 4) and to situate the paper's contribution relative to a simpler correlational analysis. In earlier exploratory work, I estimated a series of Poisson fixed-effects models regressing conflict outcomes on (uninstrumented) fire intensity at several different spatial and temporal resolutions, with no attempt to address the endogeneity of local fire activity.

Table A2 summarizes five such specifications. The first models a province-week panel of population exposure on the (unlogged) count of local fire detections and its one-week lag; the second aggregates to province-month and uses log total FRP as the regressor for conflict event counts; the third and fourth estimate kabupaten-week and kecamatan-week panels of conflict events on log FRP with a 16-week distributed-lag structure; and the fifth repeats the kabupaten-week specification on a grid of equal-area cells rather than administrative units.

Table A2: Preliminary Poisson FE models, uninstrumented fire intensity

Specification	Unit	Outcome	Regressor	Coef. (contemp.)	p-value
Province-week	province × week	pop. exposure	n_fires	−0.0002	0.070
Province-month	province × month	events	log total FRP	0.0372	0.535
Kabupaten-week	district × week	events	log FRP	−0.0100	0.631
Kecamatan-week	sub-district × week	events	log FRP	≈ 0.0000	1.000
Equal-area grid-week	1° cell × week	events	log FRP	−0.0119	0.187

Notes: All models include unit and time fixed effects and cluster-robust standard errors; the kabupaten-, kecamatan-, and grid-week specifications additionally include 16 weekly lags of log FRP (not shown). No instrumentation for fire intensity is used in any specification. "Contemp." refers to the coefficient on the contemporaneous fire-intensity regressor.

The contemporaneous coefficient on fire intensity is statistically insignificant ($p > 0.05$) in every specification, and its sign is unstable across resolutions (negative in three of five, positive in one, and essentially zero in another). In the distributed-lag specifications (kabupaten-, kecamatan-, and grid-week), a handful of individual lag coefficients out of 17 are statistically significant at conventional levels, but these do not follow a consistent or interpretable temporal pattern and differ across specifications – the signature of multiple testing under the null rather than a real dynamic effect.

Overall, these results suggest that a purely correlational approach, regressing conflict on local fire intensity however finely resolved spatially or temporally, does not reveal a stable relationship, whether because no such relationship exists or because local fire intensity is confounded with other local determinants of conflict. This motivated the instrumented design developed in Section 4, which (i) isolates plausibly exogenous variation in local fire exposure using the physical transport of smoke, and (ii) as shown in Section 5, reveals that the average relationship is indeed null, but that an economically meaningful and far more stable relationship exists within the subset of conflict-active districts once endogeneity and zero-inflation are properly addressed.

C Additional Robustness Checks

This appendix reports, in full, the four additional robustness and specification checks summarized in Section 5.6: a Poisson IV control-function specification, an analysis of conflict-related fatalities, two checks of the instrument's exclusion restriction (a placebo test and a shared-seasonality test) plus Conley spatial-correlation standard errors, and a fully-instrumented distributed-lag IV specification.

C.1 Robustness: Poisson IV control function

As a final robustness check on the full-sample null (Table 2), I consider whether the linear IV specification is simply ill-suited to the severely zero-inflated count outcomes. Following Lu and Yamazaki (2023), who study a similarly zero-inflated count outcome (conflict events in $1^\circ \times 1^\circ$ ocean grid cells, 90% zeros) and find that a Poisson IV control-function approach (Terza et al., 2008) recovers economically meaningful effects that a linear IV does not, I implement the analogous two-step procedure on the full district panel:

1. **Stage 1:** Regress logFRP_{dt} on the excluded instruments $(\text{logZ}_{dt}, \text{logZ}_{d,t-1})$ *and* the exogenous lag control $\text{logFRP}_{d,t-1}$, with district and year-month fixed effects, and compute the residual $\hat{v}_{dt} = \text{logFRP}_{dt} - \widehat{\text{logFRP}}_{dt}$. Including the exogenous lag control in the first stage (rather than only the excluded instruments) ensures $\hat{v}_{dt}$ is orthogonal to $\text{logFRP}_{d,t-1}$, as a correctly-specified two-stage residual inclusion estimator requires.

2. **Stage 2:** Estimate a Poisson fixed-effects model of the conflict outcome on logFRP_{dt}, $\text{logFRP}_{d,t-1}$, and the control function $\hat{v}_{dt}$, with district and year-month fixed effects. A statistically significant coefficient on $\hat{v}_{dt}$ would indicate that logFRP_{dt} is endogenous and that the Poisson estimate of its coefficient should be interpreted with caution; an insignificant coefficient indicates limited endogeneity bias.

Table A3: Poisson IV control function, full panel

	events	pv_events
logFRP_{dt}	−0.0297	−0.0133
	(0.0183)	(0.0434)
	$[p = 0.103]$	$[p = 0.759]$
$\text{logFRP}_{d,t-1}$	0.0045	0.0064
	(0.0100)	(0.0231)
	$[p = 0.652]$	$[p = 0.780]$
$\hat{v}_{dt}$ (control function)	0.0219	0.0397
	(0.0187)	(0.0446)
	$[p = 0.242]$	$[p = 0.374]$
District FE	Yes	Yes
Year-month FE	Yes	Yes
Observations	55,875	44,500

Notes: Stage 1 regresses logFRP_{dt} on the excluded instruments and the exogenous lag control $\text{logFRP}_{d,t-1}$ (with district and year-month fixed effects); $\hat{v}_{dt}$ is the resulting residual. Stage 2 is a Poisson fixed-effects regression. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

The control-function residual $\hat{v}_{dt}$ is statistically insignificant in both the `events` ($p = 0.242$) and `pv_events` ($p = 0.374$) Poisson models, indicating that endogeneity bias in the full-sample relationship between local fire intensity and conflict is minimal. The Poisson coefficient on logFRP_{dt} itself also remains statistically insignificant in both outcomes, whereas in Lu and Yamazaki (2023) the positive effects found with linear IV carry over to the Poisson control-function specification. I therefore conclude that the full-sample null is robust to (i) moving from a linear to a count-data (Poisson) specification, and (ii) explicitly controlling for first-stage endogeneity via the control-function approach. This strengthens the interpretation of the conflict-active heterogeneity in Section 5.3 as a genuine feature of the data, a *local* average treatment effect concentrated among districts where conflict is a meaningful, recurring outcome, rather than an artifact of model misspecification in the full sample.

C.2 Conflict severity: fatalities

The outcomes considered so far (`events`, `pv_events`) measure conflict *intensity* in terms of event counts. A natural complementary outcome is conflict *severity*: the number of fatalities ACLED records for each district-month, `fatalities`. This outcome is even more extremely zero-inflated than `events`: 98.9% of district-months record zero fatalities, with a mean of

0.028 fatalities per district-month, so I report the same set of specifications used above: the full-panel linear IV (distributed-lag and contemporaneous-only), the $\tau \geq 30\%$ conflict-active subsample, and the Poisson IV control function. Table A4 reports the results.

Table A4: Fatalities: IV estimates across specifications

Specification	logFRP_{dt}	$\hat{v}_{dt}$	Observations
Full panel, distributed lags (IV-5)	-0.0223 (0.0166) [$p = 0.189$]	–	55,875
Full panel, contemporaneous (IV-6)	-0.0135 (0.0097) [$p = 0.176$]	–	55,875
Conflict-active, $\tau \geq 30\%$	-0.1632 (0.1124) [$p = 0.156$]	–	10,750
Poisson IV control function (full panel)	-0.8832^{***} (0.0599) [$p < 0.001$]	$+0.7868^{***}$ (0.0613) [$p < 0.001$]	20,496

Notes: "Full panel, distributed lags" and "contemporaneous" mirror IV-1/IV-3 in Table 2 but with `fatalities` as the outcome. "Conflict-active" mirrors the $\tau \geq 30\%$ row of Table 3. The Poisson IV control function mirrors Table A3; its observation count is far lower because `fepois` drops district $\times$ year-month fixed-effect groups with no within-group variation in `fatalities`, which are extremely common given 98.9% zeros. Standard errors clustered by province in parentheses (Poisson model uses iid SEs). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

The linear IV estimates for fatalities are qualitatively similar in sign to the `events` and `pv_events` results: negative in all three linear specifications, including the $\tau \geq 30\%$ conflict-active subsample, where the point estimate (-0.163) is large relative to the mean, but none reach conventional significance levels ($p \in [0.156, 0.189]$). This is unsurprising given that fatalities are roughly an order of magnitude rarer than events of any kind (98.9% vs. 82.5% zeros), so the linear IV likely lacks power to detect an effect of similar relative magnitude to the event-count results.

The Poisson IV control function tells a different, and more cautionary, story. Both logFRP_{dt} (-0.883, $p < 0.001$) and the control-function residual $\hat{v}_{dt}$ ($+0.787$, $p < 0.001$) are highly significant, in sharp contrast to the `events` and `pv_events` Poisson models in Table A3, where $\hat{v}_{dt}$ was insignificant. A significant $\hat{v}_{dt}$ indicates that logFRP_{dt} is endogenous with respect to fatalities even after instrumenting, so the Poisson coefficient on logFRP_{dt} in this specification should *not* be read as a causal estimate. Compounding this, the sample used by `fepois` drops to 20,496 observations (37% of the full panel), because Poisson's conditional MLE drops any district $\times$ year-month fixed-effect group with no within-group variation in the outcome, which is overwhelmingly common when 98.9% of observations are zero. This non-random sample restriction further undermines a causal interpretation of the Poisson IV fatalities coefficient.

Altogether, I do not find robust evidence that wildfire exposure causally affects conflict-related fatalities: the linear IV estimates are suggestively negative but imprecise, and the one specification that returns a large, precisely-estimated effect (Poisson IV control function)

shows clear signs of residual endogeneity and a severely restricted, non-representative sample. This is itself an informative null: it suggests that the effect documented in Sections 5.4–5.5 operates primarily on the *occurrence* of low-fatality events such as riots, rather than on the occurrence or severity of lethal violence, consistent with the event-type decompositions, where the effect was concentrated in riots and violence against civilians and absent for battles and organized violence. A clean test of effects on fatalities would likely require a longer panel or a coarser unit of aggregation to overcome the extreme rarity of fatal events.

C.3 Exclusion-restriction checks: a placebo test and a shared-seasonality test

The upwind instrument's validity rests on an exclusion restriction: upwind fire activity should affect a district's conflict outcomes only through its effect on that district's own local fire/haze exposure, not through some other channel. I implement two checks of this assumption, plus a check on the standard-error assumptions used for inference throughout the paper.

Placebo test: instrument assigned from a distant, non-adjacent district. Each district is randomly paired with a "placebo partner" district located at least 500 km away, well beyond the 300 km upwind search radius used to construct the real instrument, and that partner's upwind-instrument series is substituted in as a placebo regressor. Smoke from a randomly-assigned district roughly 1,800 km away on average (range 507–4,917 km) has no physical transport channel to the focal district, so if the exclusion restriction holds, this placebo instrument should show no relationship with the focal district's conflict outcomes. Table A5 reports reduced-form regressions of conflict on the real and placebo instruments in the $\tau \geq 30\%$ conflict-active subsample, where the real effect is significant.

Table A5: Placebo test: real vs. distant-district instrument, $\tau \geq 30\%$ conflict-active sample

	events	pv_events
Real upwind instrument	-0.0230^{***} (0.0062) [$p = 0.001$]	-0.0095^{**} (0.0036) [$p = 0.013$]
Placebo (distant) instrument	$+0.0200$ (0.0130) [$p = 0.134$]	$+0.0026$ (0.0034) [$p = 0.456$]

Notes: "Real upwind instrument" regresses the outcome on $\log Z_{dt}$ and its one-month lag for the focal district. "Placebo (distant) instrument" substitutes the corresponding series from a randomly-assigned partner district at least 500 km away. Both rows include district and year-month fixed effects; standard errors clustered by province. $N = 10{,}750$ in both rows. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

The contrast is sharp: the real instrument reproduces the negative, statistically significant

relationship documented in Section 5.3, while the placebo instrument is small, wrong-signed (positive), and statistically insignificant for both outcomes. Consistent with this, the placebo instrument is also a far weaker predictor of the focal district's own fire intensity (effective first-stage $F = 0.3$) than the real instrument is at the same threshold (effective $F = 48.3$, Table 3). This supports the claim that the effect identified in Section 5.3 is specifically attributable to wind-transported smoke from genuinely upwind locations, rather than to some spurious correlation that any sufficiently large fire signal elsewhere in the country would produce.

Shared regional seasonal cycles. A subtler threat to the exclusion restriction is that upwind and downwind districts may share a common seasonal land-use calendar, for example a province-wide dry-season agricultural-burning cycle, so that the upwind instrument could partly be picking up regional seasonality rather than purely exogenous, wind-driven smoke transport. The year-month fixed effects in the main specification absorb national-level seasonality, but not seasonality specific to a province that recurs at the same calendar month every year. I add a province $\times$ calendar-month fixed effect to both stages, which absorbs any province-specific seasonal cycle shared by upwind and downwind districts within that province. Roughly 47% of the variance in the raw upwind instrument is explained by province $\times$ calendar-month fixed effects alone, confirming that fire activity is, unsurprisingly, seasonal and regional; the remaining 53% is the idiosyncratic, month-to-month wind-direction variation the design relies on for identification. Table A6 reports the second-stage estimates with this additional fixed effect.

Table A6: Robustness to province $\times$ calendar-month fixed effects

	Full panel		$\tau \geq 30\%$ conflict-active	
	events	pv_events	events	pv_events
logFRP_{dt}	$+0.0135$ (0.0124)	-0.0036 (0.0035)	-0.1077 (0.0797)	-0.0864^{***} (0.0228)
	$[p = 0.282]$	$[p = 0.311]$	$[p = 0.186]$	$[p < 0.001]$
Observations	55,875	55,875	10,750	10,750

Notes: All specifications add a province $\times$ calendar-month fixed effect to the district and year-month fixed effects used elsewhere in the paper. With this added fixed effect, the effective first-stage F (Montiel Olea and Pflueger, 2013) is 88.8 in the full panel (82.6 at baseline) and 20.1 in the $\tau \geq 30\%$ subsample (48.3 at baseline). Standard errors clustered by province in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

In the full panel, the instrument retains its relevance after absorbing province-level sea-

sonality (effective $F = 88.8$). In the $\tau \geq 30\%$ subsample, it weakens noticeably (effective $F = 20.1$): this is above the conventional threshold of 10 and the conservative 5% critical value for a 20% worst-case bias (15.1), but below the corresponding value for a 10% worst-case bias (23.1), so the conflict-active estimates in this table carry somewhat more weak-instrument risk than the baseline estimates. The $\tau \geq 30\%$ conflict-active result for `pv_events` is essentially unchanged, and if anything slightly larger (-0.086 vs. -0.046 at baseline) and remains highly significant. The `events` result, however, is more sensitive to this specification: the point estimate is similar in sign and magnitude to the baseline (-0.108 vs. -0.112), but loses statistical significance ($p = 0.186$ vs. $p = 0.005$ at baseline). I read this honestly as a genuine, if partial, qualification rather than a result to explain away: part of the baseline `events` effect may co-move with province-level seasonal fire activity that the year-month fixed effects alone do not fully absorb, whereas the political-violence-specific result is robust to this stricter specification. Combined with the placebo test above, these checks support the core finding – the heterogeneous effect concentrated in riots and violence against civilians documented in Sections 5.3–5.4 is not an artifact of a weak exclusion restriction, though the precision of the all-events result in particular should be read with this caveat in mind.

Conley spatial-correlation standard errors. The province-clustered standard errors used throughout this paper allow for arbitrary correlation *within* a province but assume independence *across* provinces. Since the upwind instrument is built from smoothly-varying wind fields, residual spatial correlation between geographically close districts in different provinces is plausible and would not be captured by province clustering. I therefore also compute Conley (Conley, 1999) spatial-HAC standard errors, which allow residual correlation between any two districts within a chosen distance cutoff regardless of province, for the same month. `pyfixest` does not implement Conley standard errors for IV models, so I implement the estimator directly as a two-way-demeaned 2SLS/GMM sandwich, with the spatial-HAC "meat" computed using a uniform kernel over the cutoff distance; only cross-sectional, same-month spatial correlation is modeled, since the year-month fixed effects already absorb common time trends. As a validation check, this estimator exactly reproduces the point estimates reported in Tables 3 and 4, and its province-clustered standard errors are within rounding of `pyfixest`'s reported CRV1 standard errors (e.g., 0.0361 vs. 0.0369 for the `events` coefficient at $\tau \geq 30\%$, the small remaining gap attributable to finite-sample degrees-of-freedom corrections). Table A7 reports Conley standard errors at three distance cutoffs (100, 200, and 400 km) for both headline result tables.

Table A7: Conley spatial-correlation standard errors, $\tau \geq 30\%$ conflict-active sample

Outcome	Coef. (logFRP_{dt})	Cluster SE (province)	Conley SE by cutoff		
			100 km	200 km	400 km
events	−0.1117	0.0369 [$p = 0.005$]	0.0637 [$p = 0.079$]	0.0643 [$p = 0.083$]	0.0647 [$p = 0.085$]
pv_events	−0.0458	0.0156 [$p = 0.006$]	0.0291 [$p = 0.115$]	0.0301 [$p = 0.127$]	0.0305 [$p = 0.133$]
Riots/VAC	−0.0462	0.0134 [$p = 0.002$]	0.0192 [$p = 0.016$]	0.0188 [$p = 0.014$]	0.0188 [$p = 0.014$]
Battles/Explosions	+0.0003	0.0068 [$p = 0.964$]	0.0162 [$p = 0.985$]	0.0172 [$p = 0.986$]	0.0172 [$p = 0.986$]

Notes: Coefficients are identical across columns by construction (Conley correction affects only standard errors, not point estimates). "events" and "pv_events" rows correspond to Table 3, $\tau \geq 30\%$; "Riots/VAC" and "Battles/Explosions" rows correspond to Table 4. p-values computed from a normal approximation. $N = 10{,}750$ for all rows.

Conley standard errors are uniformly larger than the province-clustered standard errors, by a factor of roughly 1.4 to 2.5 depending on the outcome, consistent with genuine residual spatial correlation that crosses province boundaries and that province clustering alone does not capture. The consequences for inference differ sharply by outcome. Under Conley standard errors, neither the all-events nor the political-violence result in Table 3 is significant at the 5% level: at the 200 km cutoff, the all-events estimate is significant only at the 10% level ($p = 0.08$), and the political-violence estimate is not significant at conventional levels ($p = 0.13$). I therefore no longer regard those two results as robust to this stricter, and arguably more appropriate, treatment of spatial dependence. The Riots/VAC result in Table 4, however, remains significant at the 5% level under every cutoff considered ($p \approx 0.014$–0.016), and the null result for Battles/Explosions remains a precise zero throughout. Because the Riots/VAC result is also the one carrying the paper's central mechanism claim (Section 5.4), this is the most important result to have survived the stricter inference; the broader `events` and `pv_events` results in Table 3 should now be read as suggestive rather than conclusively significant, and I revise the headline framing in Section 6 accordingly.

C.4 Robustness: fully-instrumented distributed-lag IV

As noted in Section 4, the main distributed-lag specifications instrument only the contemporaneous treatment, logFRP_{dt}, because `pyfixest` supports a single endogenous regressor; the one-month lag, $\text{logFRP}_{d,t-1}$, enters as an uninstrumented control. If lagged fire intensity is itself endogenous, in the same way the contemporaneous term is, this could in principle contaminate the estimate. I address this directly by instrumenting *both* logFRP_{dt} and $\text{logFRP}_{d,t-1}$ simultaneously, using both logZ_{dt} and $\text{logZ}_{d,t-1}$ as instruments (an exactly-identified two-endogenous, two-instrument 2SLS system), implemented with the same two-way-demeaned GMM estimator validated against `pyfixest` in Section C.3. Table A8 reports

the results alongside the partially-instrumented baseline.

Table A8: Fully-instrumented vs. partially-instrumented distributed-lag IV

	Partially instrumented (baseline)		Fully instrumented	
Specification	logFRP_{dt}	$\text{logFRP}_{d,t-1}$	logFRP_{dt}	$\text{logFRP}_{d,t-1}$
Full panel, events	0.0008 [$p = 0.906$]	0.0003 [$p = 0.947$]	-0.0128 [$p = 0.111$]	0.0290^{**} [$p = 0.019$]
Full panel, pv_events	-0.0021 [$p = 0.314$]	0.0017 [$p = 0.138$]	-0.0064^{**} [$p = 0.017$]	0.0107^{***} [$p = 0.006$]
$\tau \geq 30\%$, events	-0.1117^{***} [$p = 0.005$]	–	-0.1140^{***} [$p = 0.003$]	0.0245 [$p = 0.633$]
$\tau \geq 30\%$, pv_events	-0.0458^{***} [$p = 0.006$]	–	-0.0607^{***} [$p = 0.001$]	0.0468^{***} [$p = 0.004$]
$\tau \geq 30\%$, Riots/VAC	-0.0462^{***} [$p = 0.002$]	–	-0.0654^{***} [$p < 0.001$]	0.0565^{***} [$p = 0.001$]
$\tau \geq 30\%$, Battles/Explosions	0.0003 [$p = 0.964$]	–	0.0048 [$p = 0.525$]	-0.0097 [$p = 0.189$]

Notes: "Partially instrumented" reproduces the baseline distributed-lag coefficients from Tables 2, 3, and 4 ($\text{logFRP}_{d,t-1}$ omitted for the $\tau \geq 30\%$ rows, which originally entered as a control whose coefficient was not separately tabulated). "Fully instrumented" uses both logZ_{dt} and $\text{logZ}_{d,t-1}$ as instruments for both logFRP_{dt} and $\text{logFRP}_{d,t-1}$ simultaneously. Standard errors clustered by province. $N = 55{,}875$ for full-panel rows, $N = 10{,}750$ for $\tau \geq 30\%$ rows. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

This check reinforces the paper's central conflict-active and event-type results, and if anything strengthens them: the contemporaneous coefficient on `events` moves from -0.112 to -0.114, on `pv_events` from -0.046 to -0.061, and on Riots/VAC from -0.046 to -0.065, all remaining significant at the 1% level, while Battles/Explosions remains an insignificant near-zero. The uninstrumented-lag simplification disclosed in Section 4 was not driving these results.

It also complicates the full-panel null, however: properly instrumenting the lag reveals a statistically significant negative contemporaneous effect on `pv_events` in the full panel (-0.0064, $p = 0.017$), where the baseline partially-instrumented specification found a null ($p = 0.314$); the `events` coefficient remains insignificant but moves from essentially zero to a marginally negative -0.013 ($p = 0.111$). In addition, the lagged coefficient, which was small and largely insignificant under partial instrumentation, becomes economically large and highly significant and positive once properly instrumented, in every specification except the $\tau \geq 30\%$ `events` row. The pattern that emerges, a negative contemporaneous effect followed by a positive one-month-lagged effect, is suggestive of a short-run disruption followed by a partial rebound in conflict activity the following month, rather than a pure suppression. This is a new pattern not visible under the baseline specification, and I treat it as a suggestive avenue for future work (Section 7) rather than a settled finding, since it emerges from a single robustness check rather than a result independently replicated across the other checks in this section.

References

Malvina Bondy, Sefi Roth, and Lutz Sager. Crime is in the air: The contemporaneous relationship between air pollution and crime. *Journal of the Association of Environmental and Resource Economists*, 7(3):555–585, 2020.

Marshall Burke, Solomon M. Hsiang, and Edward Miguel. Climate and conflict. *Annual Review of Economics*, 7(1):577–617, 2015.

Jesse Burkhardt, Jude Bayham, Ander Wilson, Ellison Carter, Jesse D. Berman, Katelyn O'Dell, Bonne Ford, Emily V. Fischer, and Jeffrey R. Pierce. The effect of pollution on crime: Evidence from data on particulate matter and ozone. *Journal of Environmental Economics and Management*, 98:102267, 2019.

T. G. Conley. Gmm estimation with cross sectional dependence. *Journal of Econometrics*, 92(1):1–45, 1999.

Evan Herrnstadt, Anthony Heyes, Erich Muehlegger, and Soodeh Saberian. Air pollution and criminal activity: Microgeographic evidence from chicago. *American Economic Journal: Applied Economics*, 13(4):70–100, 2021.

H. Hersbach, B. Bell, P. Berrisford, G. Biavati, A. Horányi, J. Muñoz Sabater, J. Nicolas, C. Peubey, R. Radu, I. Rozum, D. Schepers, A. Simmons, C. Soci, D. Dee, and J-N. Thépaut. Era5 monthly averaged data on single levels from 1940 to present. Copernicus Climate Change Service (C3S) Climate Data Store (CDS), 2023.

Solomon M. Hsiang, Kyle C. Meng, and Mark A. Cane. Civil conflicts are associated with the global climate. *Nature*, 476(7361):438–441, 2011.

Solomon M. Hsiang, Marshall Burke, and Edward Miguel. Quantifying the influence of climate on human conflict. *Science*, 341(6151):1235367, 2013.

Seema Jayachandran. Air quality and early-life mortality: Evidence from indonesia's wildfires. *Journal of Human Resources*, 44(4):916–954, 2009.

Yifan Lu and Satoshi Yamazaki. Fish to fight: Does catching more fish increase conflicts in indonesia? *World Development*, 170:106337, 2023.

Miriam E. Marlier, Ruth S. DeFries, Apostolos Voulgarakis, Patrick L. Kinney, James T. Randerson, Drew T. Shindell, Yang Chen, and Greg Faluvegi. El niño and health risks from landscape fire emissions in southeast asia. *Nature Climate Change*, 3:131–136, 2013.

Edward Miguel, Shanker Satyanath, and Ernest Sergenti. Economic shocks and civil conflict: An instrumental variables approach. *Journal of Political Economy*, 112(4):725–753, 2004.

José L. Montiel Olea and Carolin Pflueger. A robust test for weak instruments. *Journal of Business & Economic Statistics*, 31(3):358–369, 2013.

Tamara L. Sheldon and Chandini Sankaran. The impact of indonesian forest fires on singaporean pollution and health. *American Economic Review: Papers & Proceedings*, 107(5): 526–529, 2017.

Douglas Staiger and James H. Stock. Instrumental variables regression with weak instruments. *Econometrica*, 65(3):557–586, 1997.

Joseph V. Terza, Anirban Basu, and Paul J. Rathouz. Two-stage residual inclusion estimation: Addressing endogeneity in health econometric modeling. *Journal of Health Economics*, 27(3):531–543, 2008.